NATHISHWAR C

Generative AI Engineer | AI Applications Engineer | LLM & RAG Developer | FastAPI | Python
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Portfolio | Linked in | Github

SUMMARY

AI & Machine Learning Engineering student specializing in **Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, and scalable AI applications**. Experienced in designing production-ready AI systems using **Python, FastAPI, LangChain, Hugging Face, PostgreSQL, Redis, Docker, and FAISS**. Skilled in building intelligent document search, AI tutoring systems, semantic retrieval pipelines, and automation workflows. Passionate about solving real-world business problems through modern AI engineering, scalable backend architecture, and production-quality software development.

SKILLS SUMMARY

Programming: Python, SQL, JavaScript , TypeScript

Artificial Intelligence : Machine Learning, Deep Learning, Neural Networks, Feature Engineering, Model Evaluation, Data Preprocessing ,Model Deployment, Scikit-Learn, PyTorch

Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG),AI Agents,Prompt Engineering, Prompt Chaining, Context Engineering ,Semantic Search,Embeddings,Vector Search,Document Intelligence, Knowledge Retrieval, LLM Evaluation, Hallucination Reduction,OpenAI API,Google Gemini API,Hugging Face Transformers,LangChain,LangGraph

Backend: FastAPI, REST APIs , Python , Authentication , JWT , Microservices

Databases: PostgreSQL, Redis, FAISS , Vector DB

DevOps: Docker, Linux, CI/CD , Github actions , Vercel , Render , AWS (Basic)

Tools: Git, GitHub, Postman, VS Code

PROJECTS

KnowGrow AI — Adaptive Learning Intelligence Platform

• Architected a production-ready AI learning platform integrating **LLMs, Retrieval-Augmented Generation (RAG), AI tutoring, and semantic search** for personalized education.

• Designed a modular FastAPI backend supporting authentication, AI tutoring, course management, quizzes, revision tracking, and progress analytics.

• Built an adaptive recommendation engine using confidence-based spaced repetition algorithms and personalized learning intelligence.

• Integrated vector search using FAISS embeddings to retrieve relevant educational content with low-latency semantic retrieval.

• Engineered AI-powered content generation workflows capable of producing quizzes, explanations, summaries, and learning paths.

• Containerized backend services using Docker with PostgreSQL and Redis to support scalable deployment.

• Designed RESTful APIs following API-first architecture for future web and mobile integration.

AI Knowledge Assistant (RAG-based)

• Developed an enterprise Retrieval-Augmented Generation (RAG) application enabling intelligent document-based question answering.

• Built automated document ingestion pipelines supporting parsing, chunking, embedding generation, indexing, and semantic retrieval.

• Implemented vector similarity search using FAISS and Hugging Face embedding models to improve contextual document retrieval.

• Connected LLM inference with retrieval pipelines to reduce hallucinations and improve factual accuracy.

• Designed scalable FastAPI APIs supporting multi-document querying, conversational memory, and AI-assisted knowledge retrieval.

• Structured the application following modular service architecture for future enterprise deployment.

AI Resume Intelligence Platform

. • Developed an AI-powered resume intelligence platform capable of ATS scoring, semantic job matching, and personalized resume optimization.

• Implemented embedding-based similarity search using Sentence Transformers and vector databases.

• Built multi-agent workflows for resume analysis, job description parsing, and skill-gap detection.

• Integrated LLMs to generate recruiter-style feedback, interview preparation guidance, and resume rewriting suggestions.

• Designed modular APIs enabling scalable AI inference and future SaaS deployment.

WORK EXPERIENCE

AI/ML Intern — Google Developer Student Club (GDSC)

Oct 2024 – Dec 2024

• Engineered end-to-end Machine Learning pipelines using Scikit-Learn for predictive analytics and classification tasks.

• Performed feature engineering, preprocessing, model training, evaluation, and hyperparameter optimization.

• Built reusable Python modules for data preprocessing and inference workflows.

• Collaborated in Agile development using Git-based version control and code reviews.

• Applied software engineering best practices to improve code quality and maintainability.

EDUCATION

B.Tech Artificial Intelligence and DataScience
Dhanalakshmi Srinivasan Engineering College
9.0 CGPA
Aug 2023 - May 2027

Marutham matric.Hr.Sec.School
Bio Mathematics
89.8%
Jun 2021 - May 2023

CERTIFICATES

- Machine Learning - Stanford & Deeplearning.ai
- Prompt Engineering — Google
- AI Essentials - Google
- Problem Solving — HackerRank